

Course Journey

In this train, we'll familiarize ourselves with the comprehensive structure and components of the data engineering curriculum. We will take a look at how each module will provide us with the necessary tools and knowledge to excel in our careers as data engineers.

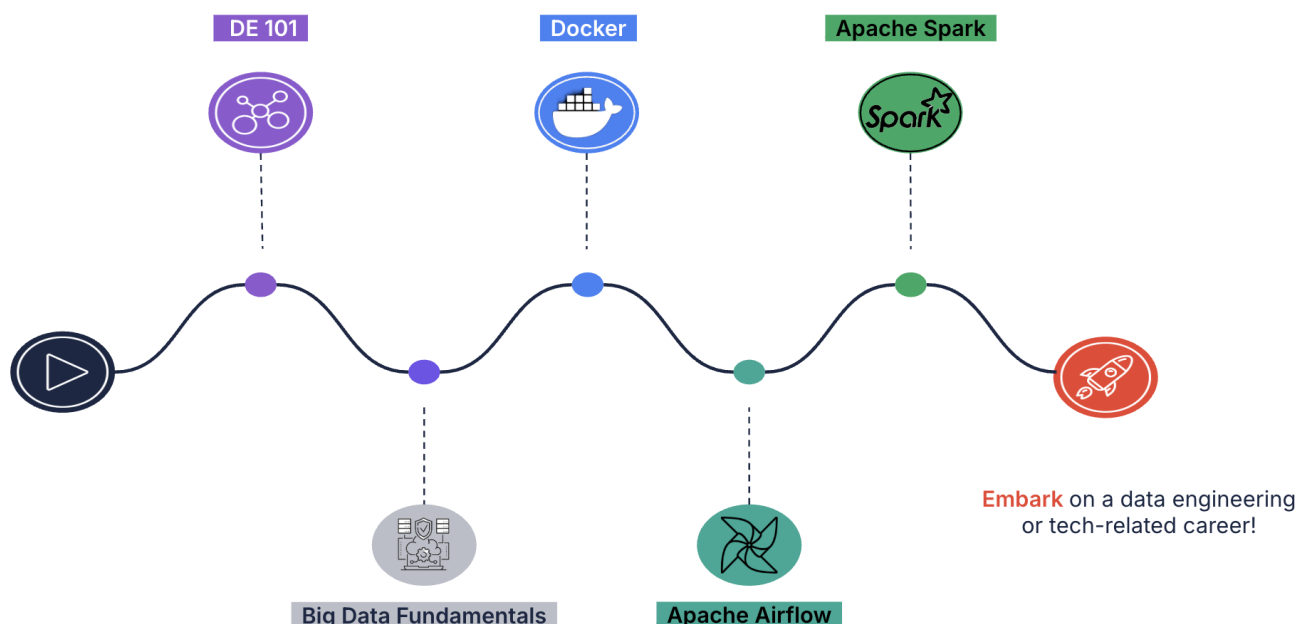
Learning Objectives

By the end of this train you should be able to:

- Gain a thorough understanding of the course's organization and its structural components.
- Acquire insights into how each part of the course contributes to developing your capabilities and skills as a data engineer.

Your Data Engineering Journey

This course is structured into five core sprints, each building on the last. Think of each sprint as a stepping stone, preparing you with the essential skills needed in the field of data engineering.



Sprint 1: DE101 – Foundations of Data Engineering

This is your launchpad – where you explore what data engineering is, why it's important, and how it supports the data lifecycle in organizations.

- **Introduction to Big Data:** Learn why data has become a valuable business asset.
- **Data Modeling and DBMS:** Understand relational databases and schema design.
- **Introduction to ETL:** Discover how data is extracted, transformed, and loaded across systems.
- **Big Data Architecture:** Visualize system design and end-to-end data flow.
- **Data Security:** Learn to safeguard data across the pipeline.
- **Data Lineage:** Trace the life of data from source to insight.
- **Data Quality:** Explore validation techniques to ensure trusted insights.

Skills gained: Systems thinking, data lifecycle awareness, schema literacy, quality & security fundamentals.

Sprint 2: Big Data Fundamentals

Now we get hands-on with the mechanics of big data – how it's stored, moved, and processed in large-scale environments.

- **Storing Big Data:** Understand distributed storage systems and formats.
- **Retrieving Data:** Learn to pull data using APIs, JDBC, and transfer protocols.
- **Streaming Data:** Dive into real-time ingestion using message brokers and cloud tools.
- **Delta Live Tables:** Implement declarative streaming pipelines.
- **Pipeline Orchestration:** Coordinate large-scale data movement with reliability and scale.
- **Processing Big Data:** Put it all together with frameworks like Spark.

ðŸŽ“ Skills gained: Real-time data streaming, cloud storage handling, data ingestion pipelines, orchestration.

Sprint 3: Working with Docker

Before data can be reliably processed and deployed, it needs a consistent environment. Docker lets you "package" your data apps and services for smooth deployment and collaboration.

- **Containers and Images:** Build, run, and manage containers.
- **Docker Networking:** Connect containers into a working system.
- **Volumes and Data Persistence:** Retain data across container lifecycles.
- **Docker Compose:** Build complex, multi-service applications.

ðŸŽ“ Skills gained: DevOps familiarity, containerization, reproducible environments, collaboration workflows.

Sprint 4: Working with Apache Airflow

Automation is critical in modern data engineering. Airflow gives you the power to define, monitor, and scale data workflows with ease.

- **Apache Airflow Basics:** Learn DAGs and scheduling.
- **Pipeline Scalability:** Handle growing workflows with performance in mind.
- **Pipeline Automation:** Trigger and manage flows automatically.
- **Processing Big Data with PySpark and Airflow:** Combine power tools for big jobs.

ðŸŽ“ Skills gained: Workflow automation, scheduling, DAG design, operational efficiency.

Sprint 5: Working with PySpark

This is where everything scales â€” Spark helps you process massive datasets efficiently. You'll learn how to write distributed processing jobs and manage performance.

- **Intro to Big Data Processing:** The "why" and "how" of distributed computing.
- **Intro to Spark:** Understand Spark's core concepts and components.
- **Spark Deeper Concepts:** Dive into partitions, transformations, and shuffles.
- **Data Engineering in Spark:** Filter, aggregate, and model data effectively.
- **Optimizing Spark:** Tune performance like a pro.
- **Data Quality with PySpark:** Build validation logic into your codebase.

ðŸŽ“ Skills gained: Distributed data processing, PySpark fluency, pipeline optimisation, data validation at scale.

Conclusion

The journey ahead is filled with practical, in-demand, and exciting skills that will shape your future as a data engineer. By the end of this course, you'll not only understand the ecosystem â€” you'll be able to design, automate, scale, and secure real-world data pipelines using best-in-class tools. Let's dive in.

Key Takeaways

- **Takeaway 1:** This course equips you with both foundational knowledge and practical tools.
- **Takeaway 2:** Each sprint builds new capabilities in storage, processing, automation, and architecture.
- **Takeaway 3:** You'll master essential tools like Docker, Airflow, and PySpark.
- **Takeaway 4:** You'll learn to think like a systems designer â€” integrating tools for maximum efficiency.
- **Takeaway 5:** By the end, you'll be ready to contribute as a professional data engineer in real teams.